# PV Keyboard User Guide

## Revision History

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|---|---|---|
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# Overview

This document describes the features and usage of the Photovoltaic (PV) Bluetooth LE Keyboard that is enabled by Atmosic's ATM3 energy harvesting Bluetooth LE solution. The keyboard is designed around the ATM3, Photovoltaic (PV) cells to harvest energy from ambient light, as well as a coin cell battery and supercapacitor for energy storage.

The Atmosic ATM3 solution combines ultra low power Bluetooth LE with an integrated Power Management Unit (PMU) that enables direct connections to energy harvesting elements like PV cells. The direct connection to the ATM3 PMU has multiple benefits, including reducing the overall cost of adding energy harvesting as well as improving the efficiency with which the harvested energy is used. In addition, the low power consumption of the ATM3 means a much smaller PV cell can be used, further reducing both the cost and area of the PV cell and enabling more streamlined industrial designs.

The PV Keyboard is designed to operate at indoor light levels primarily from harvested light and excess harvested energy stored in the supercapacitor. The CR2032 coin cell battery is only utilized when sufficient harvested and stored energy is not available, such as extended periods of operation in a dark environment. By efficiently using harvested energy as much as possible the battery life of the keyboard is greatly extended.

The PV Keyboard is provided for demonstration and evaluation purposes only and is not a commercial product for sale. A list of feature limitations and future enhancements is provided at the end of this document.

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# PV Keyboard Features

The Atmosic PV Keyboard is shown in Figure 1. A commercially available PV keyboard was used for the housing with an Atmosic designed circuit board completely replacing the internal electronics. Only half of the PV cell area used in the commercially available version is required due to the low power consumption of the Atmosic ATM3.




Figure 1 - Atmosic PV Keyboard

A simplified hardware block diagram is shown in Figure 2. The hardware design is centered around the ATM3221 device from Atmosic, with an external 4 Mb QSPI flash for Bluetooth LE and application software storage.

The PV cells from SCNE in the keyboard housing are only 17 cm x 3 cm and are capable of harvesting energy even at low indoor light level. Any excess energy is stored in a 220 mF ELNA DSK super-capacitor. While there are two PV cells on this keyboard, only one of them is used.

A CR2032 coin cell battery can be used to power the device when ambient light is unavailable and the supercapacitor is depleted. The ATM3 will always use energy from the PV cell as a primary power source and use excess harvested energy that has been stored on the supercapacitor when the amount of harvested energy is insufficient. Only when there is insufficient PV generated and stored energy will the battery be used. For consistent operation it is recommended to always have a coin cell installed in the keyboard. Note that the switch on the back of the keyboard is not functional.

The key matrix on this keyboard is 8 x 18 and is supported by the I/O pins on the ATM3221. Contact Atmosic for additional information regarding key matrix connections and configurations.

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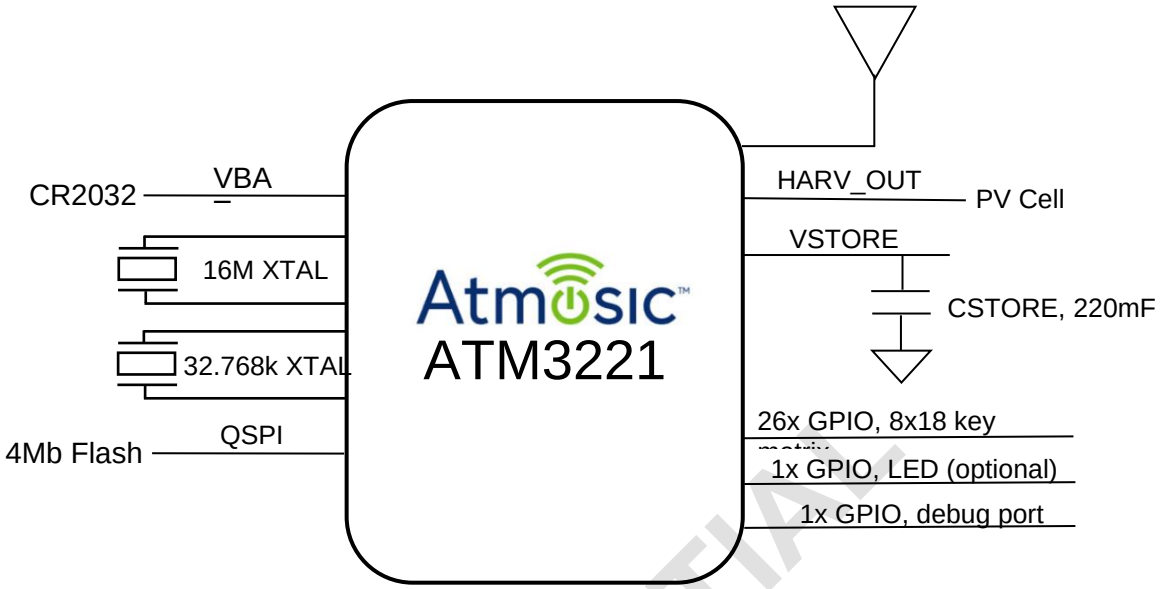



Figure 2 - PV Keyboard Block Diagram

More information about the 220 mF DSK supercapacitor can be found in the ELNA product catalog at: http://www.elna.co.jp/en/capacitor/pdf/catalog_19-20_e.pdf

# Keyboard Operation

The different operating states of the keyboard and the relationship between them are shown in Figure 3. The operations for pairing and unpairing the keyboard with a PC as well as the default configuration are described in this section.

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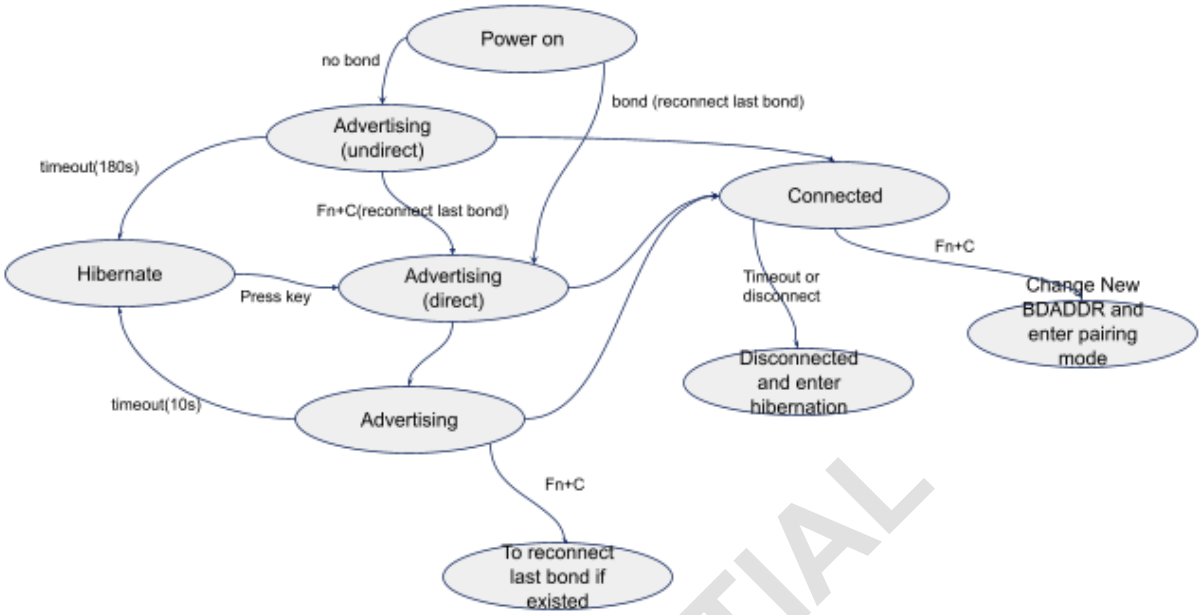



Figure 3 - Keyboard State Transitions

## Pairing and Unpairing the Keyboard

Pressing the Function key + left Shift key + Backspace key simultaneously will delete all the pairing records and enter pairing mode. At this point the keyboard is in the advertising (undirect) state shown in Figure 3 (Note that the state transitions based on Fn+left shift+backspace are not shown in the figure).

As an alternative, the Fn+c key sequence will force the keyboard which is in the connected state to change the Random Static BD Address and return to the pairing mode. If the keyboard is already in the Advertising (undirect) state, the Fn+c key sequence will reconnect to the last bonded device.

## Default Connection Parameters

By default the keyboard will operate with the following parameters:

- Connection Interval: 7.5 ms
- Slave Latency: 20
- Connection Timeout: 2.1 s

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# Power Consumption

The power consumption of the PV Keyboard in various modes without any energy harvesting is listed in Table 1 below. These measured values were taken with the PV cells and supercapacitor disconnected.

| Keyboard State | Current ($\mu$A) |
|---|---|
| Pairing | 285 |
| Connection | 7.8 |
| 4keys/1sec | 93 |
| Hibernation (Sleep) | 1.5 |
| Reconnection | 223.5 |

Table 1 - PV Keyboard Power Consumption

This information is provided to assist in the sizing of PV cells based on usage conditions.

The keyboard has two connections to aid in the evaluation of the PV keyboard's power consumption. The V_battery wire connects to the + terminal of the battery and can be connected to a power source to measure the current consumption from the battery (with the coin cell battery removed). Measuring the current consumption with the PV cells are covered and the supercapacitor in a depleted state will duplicate the values in Table 1. If the supercapacitor is charged then there will be no current drawn from V_battery.

The V_store wire can be used to measure the voltage on the supercapacitor to determine if it is in a charged or depleted state and to measure the accumulation of energy on the supercapacitor from the PV cell.

# Interoperability Testing

Interoperability testing of the keyboard against various laptop PCs, tablets, and cell phones was performed by an independent laboratory. A test report summarizing the results of this testing is available from Atmosic.

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# Feature Limitations and Future Enhancements

Note the following limitations and missing features:

1. Functional keys for Mac OS, specifically the command and option keys, have limited support in this version.
2. The LED on the keyboard is not enabled at this time.
3. Connection parameters are not optimized for MacOS so measured power consumption may be higher than the numbers given in this document.
4. The keyboard hardware does not support rechargeable batteries. Contact your Atmosic representative for more information regarding rechargeable battery usage.

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